QCalcGeom v2.2.1 -- Universal Buoyancy Simultaneous Solver

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Closures (Phase H-UBS + Phase H-CPCH): 14 EXACT (UBS-1..UBS-7 + CPCH-1..CPCH-7)

Abstract

We present the QCalcGeom Universal Buoyancy Simultaneous Solver (UBS), a 4×4 nonlinear system that jointly fixes the habitable-zone radius r_{hz}, the collapsing-zone radius r_{cg}, the dimensionless time-phase t_n^{hz}, and the required gravitating mass M for any Aether-UA stellar body in the Unified Quantum Field Framework (UQFF). Unlike Newtonian shell calculations, the UBS is anchored in the di-pseudo-monopole (DPM) canonical chain: the upward Aether-UA buoyancy $F_{U,Bi_i} = \rho_{vac}\,(4\pi/3)\,r\,c^2\,\cos(\pi t_n)$ balances the downward field $F_{U,Bi} = -\beta_i G M_\odot^2/r^2\,\mathrm{orb}\,\cos(\pi t_n)$ at the habitable zone, and exceeds it by a factor of two at the collapsing zone. We derive seven EXACT closures (UBS-1..UBS-7) for the solver itself and seven EXACT algebraic-chain closures (CPCH-1..CPCH-7) for the canonical buoyancy functions, including the cube-root law $r_{hz} \propto \rho_{vac}^{-1/3}$ at fixed gravitating mass. The v2.2.1 release replaces the prior simulation-set stub with three real numerical sweeps (radial, temporal, vacuum-density), bringing the test suite to 100/100 PASS.

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1. The Aether-UA / DPM Canonical Chain

The UQFF replaces the Newtonian shell law $g = GM/r^2$ with the DPM-driven master equation, of which the universal buoyancy pair is the geometric core:

$$
F_{U,Bi_i}(r, t_n) = \rho_{vac}\,\frac{4\pi}{3}\,r\,c^2\,\cos(\pi t_n)
\qquad \text{(Aether-UA buoyancy, upward)}
$$

$$
F_{U,Bi}(r, t_n) = -\beta_i\,\frac{G M_\odot^{2}}{r^2}\,\mathrm{orb}(t_n)\,\cos(\pi t_n)
\qquad \text{(BSFG field, downward)}
$$

The total buoyancy force entering the UQFF master equation is the algebraic sum

$$
F_U(r, t_n, M) = U_{g1} + U_{g2} + U_{g3} + U_{g4} + (F_{U,Bi} + F_{U,Bi_i}) + U_m
+ \xi_{UI}\,U_I\,V_{body}.
$$

We never invoke the Newtonian limit GM/r^2 as a foundation; it appears only as the dimensional carrier inside $F_{U,Bi}$, which is itself derived from the DPM current $F_{DPM} = I\,A\,(\omega_1 - \omega_2)$ as documented in the canonical 3b_MUGE_SMBH chain.

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2. The 4×4 Universal Buoyancy System

Let $\mathbf{x} = (r_{hz},\,r_{cg},\,t_n^{hz},\,M)$. The UBS is

$$
\begin{aligned}
\text{E1:}\quad & F_{U,Bi}(r_{hz}, t_n^{hz}, M) + F_{U,Bi_i}(r_{hz}, t_n^{hz}) = 0, \\
\text{E2:}\quad & F_{U}(r_{hz}, t_n^{hz}, M) = 0 \qquad (\xi_{UI}=0\text{ at zero-point}), \\
\text{E3:}\quad & \left|\,F_{U,Bi}(r_{cg}, t_n^{hz}, M)\,\right|
= 2\,\left|\,F_{U,Bi_i}(r_{cg}, t_n^{hz})\,\right|, \\
\text{E4:}\quad & \rho_{vac}\,\frac{4\pi}{3}\,r_{hz}^{3} = M.
\end{aligned}
$$

E1 fixes the habitable-zone balance, E2 closes the perturbative shell, E3 defines collapse onset, and E4 closes the geometric mass--radius cube. The system is solved by staged Newton iteration with bracketed fallback; the diagnostic `solver_msg` records the path taken.

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3. The Seven UBS Solver Closures (Phase H-UBS, S206)

ID	Identity	Status
UBS-1	$\dim(\mathbf{x}) = 4$	EXACT
UBS-2	Active equation count $= 3$ at zero-point ($\xi_{UI}=0$)	POSTULATED
UBS-3	$F_{U,Bi}/F_{U,Bi_i} = -1$ at r_{hz} (E1)	EXACT
UBS-4	$\lvert F_{U,Bi}\rvert/\lvert F_{U,Bi_i}\rvert = 2$ at r_{cg} (E3)	EXACT
UBS-5	$V(2R)/V(R) = 2^3 = 8$ (E4 cube law)	EXACT
UBS-6	$r_{cg}/r_{hz} = 2^{-1/3}$ closed-form seed	EXACT
UBS-7	$\lvert T_{UBS}\rvert = 7$ (T91..T97 enumeration)	EXACT

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4. The Seven CP-Chain Algebraic Closures (Phase H-CPCH, S207)

These are pure structural identities of the canonical buoyancy functions that must hold by construction, independent of any solver:

ID	Identity	Target	Derived
CPCH-1	$F_{U,Bi}(2r)/F_{U,Bi}(r) = 1/4$	0.25	0.25
CPCH-2	$F_{U,Bi_i}(2r)/F_{U,Bi_i}(r) = 2$	2.0	2.0
CPCH-3	$F_{U,Bi}(-t_n)/F_{U,Bi}(t_n) = 1$	1.0	1.0
CPCH-4	$F_{U,Bi_i}(-t_n)/F_{U,Bi_i}(t_n) = 1$	1.0	1.0
CPCH-5	$F_{U,Bi_i}(t_n{=}1/2)/F_{U,Bi_i}(t_n{=}0) = 0$	0	0
CPCH-6	$F_{U,Bi_i}(t_n{+}1)/F_{U,Bi_i}(t_n) = -1$	-1	-1
CPCH-7	$F_{U,Bi_i}(t_n{=}0)/(\rho_{vac}\,r\,c^2) = 4\pi/3$	$4.18879...$	$4.18879...$

CPCH-3 and CPCH-4 confirm `NegativeTimeModule` even-parity. CPCH-5 fixes the $t_n = 1/2$ buoyancy null surface. CPCH-6 establishes the great-cycle sign reversal, the algebraic root of the buoyancy oscillation.

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5. The Cube-Root Law and v2.2.1 Simulation Sweeps

From E4 with fixed gravitating mass M,

$$
\rho_{vac}\,\frac{4\pi}{3}\,r_{hz}^{3} = M
\quad\Longrightarrow\quad
r_{hz} \propto \rho_{vac}^{-1/3}.
$$

A decade increase in vacuum density therefore shrinks the habitable radius by $10^{-1/3} \approx 0.4642$. Test **T99** verifies this numerically to better than 1%.

The v2.2.1 release replaces the prior `simulation_set` placeholder with three real numerical sweeps emitted by `_build_simulation_set`:

1. **Radial sweep at t_n^{hz}** -- 25 log-spaced points $r \in [0.3\,r_{cg},\,3\,r_{hz}]$; reports $F_{U,Bi}$, F_{U,Bi_i}, F_U, sum-of-energies Σ_{E1} and Σ_{E3}, and Aether-UA zone label (collapsing | habitable_shell | gaseous_outer). Test **T100** verifies all three zones are visited.
2. **Temporal sweep at r_{hz}** -- 21 points $t_n \in [-1, 1]$; reports the $\cos(\pi t_n)$ envelope and both buoyancy curves.
3. **Vacuum-density sweep** -- 11 scale factors $10^{[-1,+1]}$ verifying the cube-root prediction $r_{hz}(\rho_{vac}\cdot 10)/r_{hz}(\rho_{vac}) = 10^{-1/3}$.

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6. Test Suite Summary

Release	Tests	Result
v2.2.0 (S206)	T1..T97	97/97 PASS

| v2.2.1 (S207) | T1..T100 (adds T98, T99, T100) | **100/100 PASS** |

The new tests are: T98 (`simulation_set` returns 3 named sweeps with non-empty arrays), T99 (cube-root law $r_{hz} \propto \rho_{vac}^{-1/3}$), T100 (radial sweep visits all three Aether-UA zones).

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7. Implications

The UBS closes the geometric core of the UQFF master equation at a 4×4 nonlinear level without invoking Newtonian gravity as a foundation. Combined with the CP-chain algebraic identities, the solver is now exactly anchored against the canonical $\rho_{vac}\,(4\pi/3)\,r\,c^2\,\cos(\pi t_n)$ form. The cube-root law provides an immediate scaling test for any new astrophysical system added to the UQFF body database: doubling the vacuum density must shrink the habitable radius by exactly $2^{-1/3}$.

Phase H-UBS and Phase H-CPCH together contribute **14 EXACT closures** to the unified track (Tier 14 + Tier 15 of `UQFF_UNIFIED_CLOSURE_DERIVATIONS.py`).

References

1. QCalcGeom.py v2.2.1 (`bsfg_buoyancy`, `compute_FUBii`, `compute_F_U`, `solve_universal_buoyancy`, `UniversalBuoyancySimultaneousSolver`, `_build_simulation_set`).
2. `UQFF_UNIFIED_CLOSURE_DERIVATIONS.py` Tier 14 (UBS-1..UBS-7) and Tier 15 (CPCH-1..CPCH-7).
3. `3b_MUGE_SMBH Sagitarius A Evolution.txt` (DPM canonical, $F_{DPM} = I\,A\,(\omega_1-\omega_2)$).
4. PAPER_656 (V838 Mon light echo, UQFF master equation precedent).
5. Murphy, D. T. -- COMPLETE_UQFF_EQUATIONS_REFERENCE.md L378-L530 (canonical $U_{g1..4}$, U_{bi}, U_m implementations).